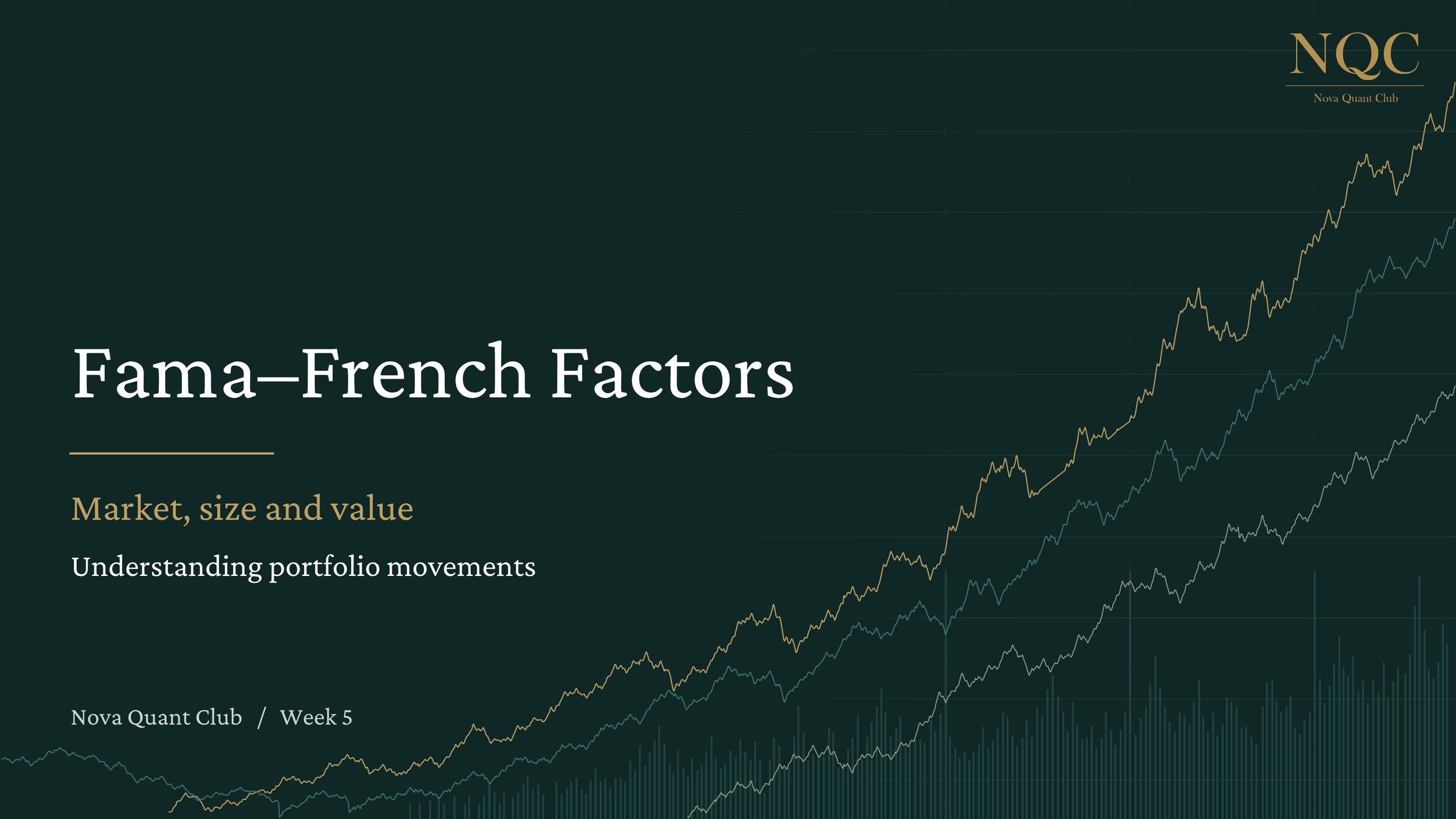
NQC
Nova Quant Club

# Fama–French Factors

## Market, size and value

Understanding portfolio movements

Nova Quant Club / Week 5



# CAPM recap

NQC
Nova Quant Club

The market as a common source of return movements

$$R_p - Rf = \alpha + \beta M\ (RM - Rf) + \varepsilon$$

**Market excess return**

The market's return above the risk-free return

**Market beta**

How strongly the portfolio moves with the market

**Residual**

The day's movement left outside the fitted return

# CAPM’s limitations

Nova Quant Club

## Market beta did not capture every average-return pattern

Market beta alone did not fully explain differences in average US stock returns

### Size

Company size helped explain average-return differences

### Book-to-market

Book-to-market helped explain average-return differences

Source: Fama & French (1992), The Journal of Finance
US stocks, 1963–1990

# Fama and French’s approach

NQC
Nova Quant Club

## Turn company characteristics into common return factors

$$R_p - Rf = \alpha + \beta M\ (RM - Rf) + \beta SMB\ SMB + \beta HML\ HML + \varepsilon$$

**Market**

Market index excess return
SPY as our proxy

**SMB**

Small-cap returns
minus big-cap returns

**HML**

High B/M – low B/M
(value – growth)

B/M = book-to-market

# SMB: Small Minus Big

A return spread between smaller and larger companies

**LONG smaller stocks**

Positive weights
sum to +1

—

**SHORT larger stocks**

Negative weights
sum to –1

# SMB: Weighting the two sides

## Long smaller stocks, short larger stocks

```
for ticker in big_caps:
    weights[ticker] = -big_caps[ticker] / total_big_caps

for ticker in small_caps:
    weights[ticker] = small_caps[ticker] / total_small_caps
```

# HML: High Minus Low

High book-to-market versus low book-to-market

**LONG value**

High book-to-market
Lower price vs book value

—

**SHORT growth**

Low book-to-market
Higher price vs book value

# HML: From data to portfolio weights

NQC
Nova Quant Club

## Get price-to-book from yfinance, then invert it

Book-to-market = 1 / price-to-book

```
stock = yf.Ticker(ticker)
mtb = stock.info.get("priceToBook")
market_cap = stock.fast_info["market_cap"]

if mtb is not None and mtb > 0 and market_cap is not None:
    btm = 1 / mtb

for ticker in value_stocks:
    weights[ticker] = value_caps[ticker] / total_value_cap

for ticker in growth_stocks:
    weights[ticker] = -growth_caps[ticker] / total_growth_cap
```

# Preparing the factor data

## Align the daily Market, SMB and HML return series

```
# Inside build_factor_data()
market = market.rename("Market")
smb = smb.rename("SMB")
hml = hml.rename("HML")

factor_data = pd.concat(
    [market, smb, hml],
    axis=1
)

factor_data = factor_data.dropna()
```

One row per date, with complete Market, SMB and HML returns

# Computing portfolio returns

NQC
Nova Quant Club

## Calculate the investment returns, then match every date

Portfolio return = sum of (weight × stock return)

```
investment_returns = compute_returns(portfolio_weights, portfolio)
investment_series = pd.Series(investment_returns, name="Portfolio")

regression_data = pd.concat(
    [investment_series, factor_data],
    axis=1
)

regression_data = regression_data.dropna().sort_index()
```

Teaching baseline: 20% in each stock · Daily risk-free return assumed zero

# Estimating the regression

NQC
Nova Quant Club

## Fit Market, SMB and HML, then read the factor exposures

```
# Inside run_regression()
factors = sm.add_constant(factors)
model = sm.OLS(returns, factors).fit()

three_factor_model = run_regression(
    regression_data["Portfolio"],
    regression_data[["Market", "SMB", "HML"]]
)

print(three_factor_model.params)
```

const: daily alpha
Market, SMB, HML: factor exposures

# Factor exposures: our portfolio

NQC
Nova Quant Club

## Using the three-factor results from our notebook

### SMB
**β = +0.009**

A +1 pp move in SMB is associated with a +0.009 pp change in our portfolio's fitted daily return.

### HML
**β = −0.446**

A +1 pp move in HML is associated with a −0.446 pp change in our portfolio's fitted daily return.

### Market
**β = +1.030**

A +1 pp market move is associated with a +1.030 pp change in our portfolio's fitted daily return.

Other factors held constant · Historical regression estimates

Equal weights · 66 daily observations · RF = 0 · Coefficients rounded

# Reading the signs: portfolio tilts

NQC
Nova Quant Club

The sign shows direction; the magnitude shows sensitivity

| SMB | HML | Market |
|---|---|---|
| $\beta > 0$<br>Small-cap tilt | $\beta > 0$<br>Value tilt | $\beta > 0$<br>Moves with the market |
| $\beta < 0$<br>Large-cap tilt | $\beta < 0$<br>Growth tilt | $\beta < 0$<br>Moves against it |

Our portfolio: almost no size tilt · growth tilt · market $\beta \approx 1.03$

Near-zero SMB/HML: little estimated style tilt. Exposures are not portfolio weights.

# HML: what does −0.446 mean?

NQC
Nova Quant Club

Direction, sensitivity and explained variation

## Sign

Negative HML exposure suggests growth-like (low B/M) behaviour.

## Magnitude

About 0.446 pp in fitted return per 1 pp change in HML.

## $R^2$

How much historical return variation the full model explains.

Growth beats value by 1 pp → HML = −1 pp

$(-0.446) \times (-1 \text{ pp}) = +0.446 \text{ pp}$

HML contribution to fitted return · Market and SMB held constant
−0.446 does not mean 44.6% of movements are explained.

# Alpha and $R^2$

Unexplained average return and historical variation explained

NQC
Nova Quant Club

## Alpha

Average daily return left unexplained by the chosen factor exposures

Positive alpha alone does not establish investment skill

## $R^2 = 0.80$

80% of historical return variation explained in this hypothetical sample

Higher sample fit does not establish better forecasts

Adding factors cannot lower ordinary $R^2$ on identical observations

# Notebook and assignment

NQC
Nova Quant Club

## Market-cap weighting and model comparison

### Live notebook

Build the factors
Align daily returns
Estimate factor exposures

### Your assignment

1. Set market-cap weights
2. Recompute portfolio returns
3. Fit CAPM and three-factor
4. Plot both regressions
5. Compare and explain $R^2$

Keep the factor portfolios and regression dates identical